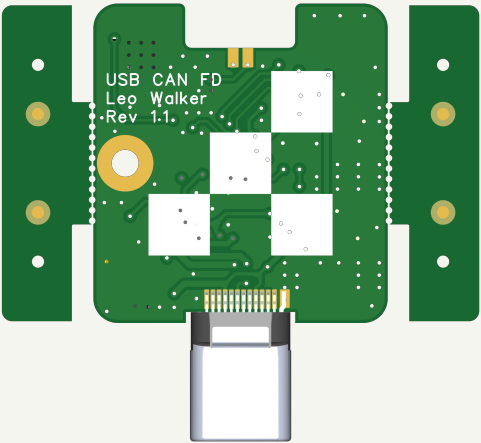
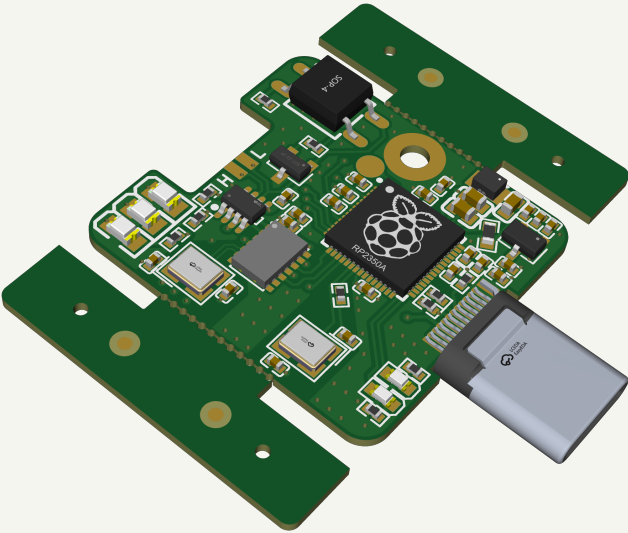
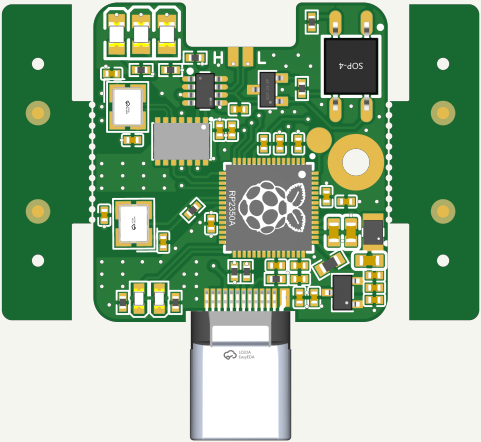


# USB CAN FD Module PCB

A USB Module which supports CAN FD communication using the RP2354A microcontroller and the MCP2518FD CAN controller.  
Used for easy CAN bus diagnostics and data logging / analysing.

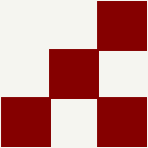
## PCB Requirements:

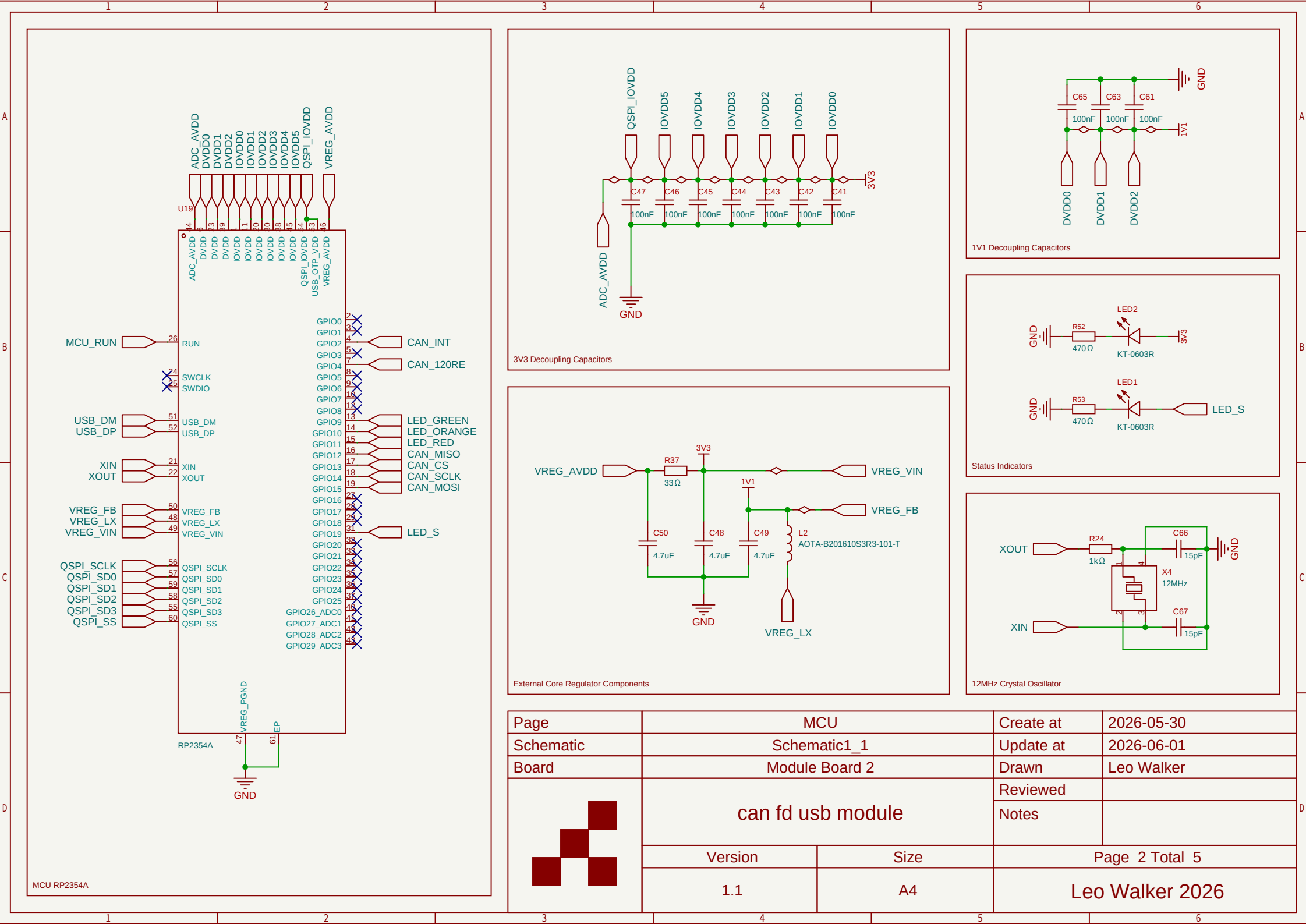
- USB Male interface
- 2-pin picoblade for CAN bus interface
- Software toggleable termination resistor
- Bus status LEDs
- Power good and activity indicators
- MCU (RP2354A)
- 8Mbps CAN FD controller and transceiver
- Framework laptop module compatible footprint.

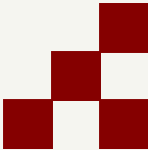


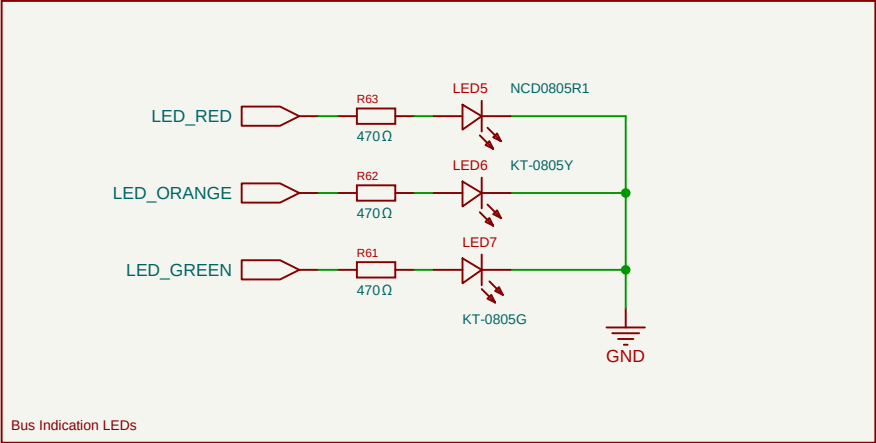
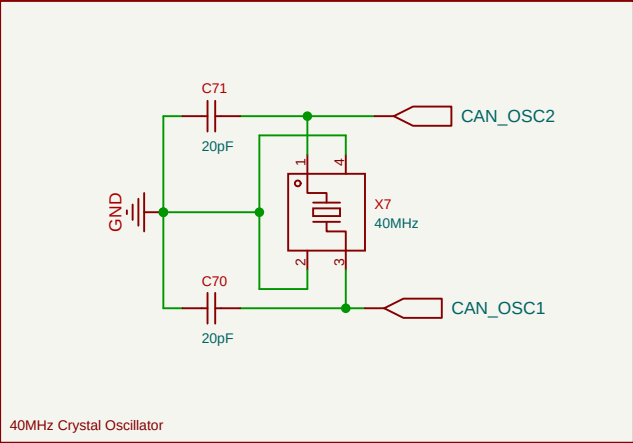
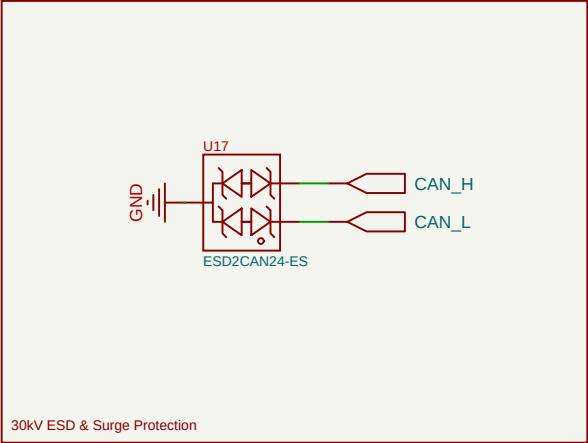
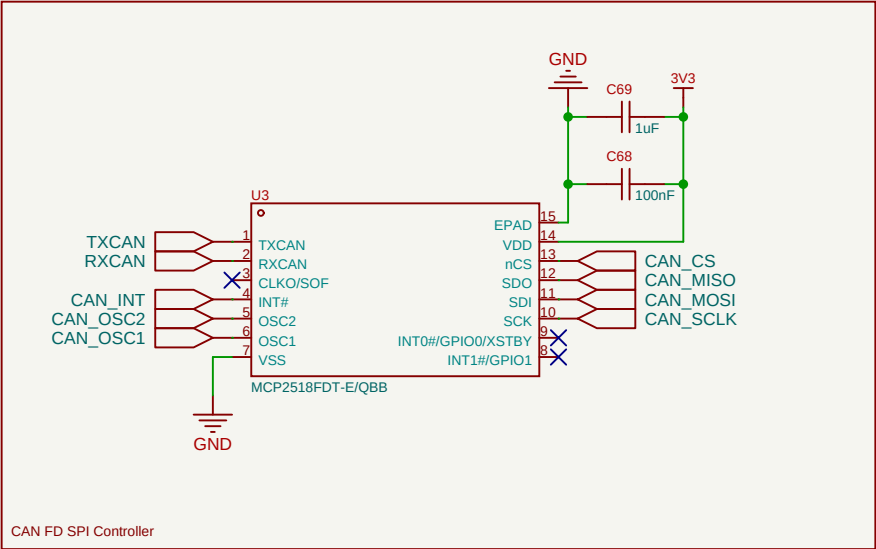
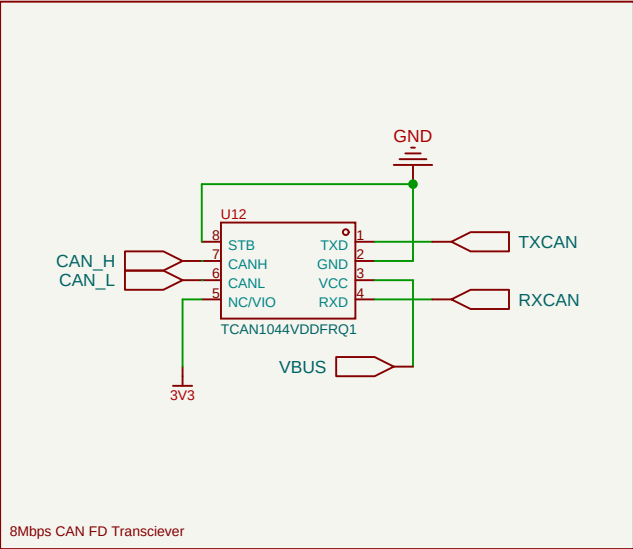
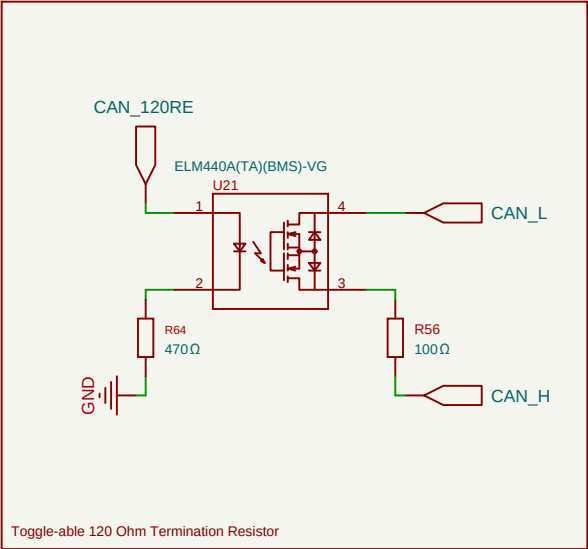
## Revision Changes:

Rev 1.1 - Added bus status indicators.  
Added board outline slot for a pico blade connector to be mounted recessed, reducing the modules height profile.  
Fixed USB CC2 issue, male connector only needs 1 5.1kOhm on CC1.

|                                                                                      |                   |      |                 |            |
|--------------------------------------------------------------------------------------|-------------------|------|-----------------|------------|
| Page                                                                                 | Cover             |      | Create at       | 2026-06-01 |
| Schematic                                                                            | Schematic1_1      |      | Update at       | 2026-06-01 |
| Board                                                                                | Module Board 2    |      | Drawn           | Leo Walker |
|  | can fd usb module |      | Reviewed        |            |
|                                                                                      |                   |      | Notes           |            |
|                                                                                      | Version           | Size | Page 1 Total 5  |            |
|                                                                                      | 1.0               | A4   | Leo Walker 2026 |            |



|                                                                                      |                   |      |                 |            |
|--------------------------------------------------------------------------------------|-------------------|------|-----------------|------------|
| Page                                                                                 | MCU               |      | Create at       | 2026-05-30 |
| Schematic                                                                            | Schematic1_1      |      | Update at       | 2026-06-01 |
| Board                                                                                | Module Board 2    |      | Drawn           | Leo Walker |
|  | can fd usb module |      | Reviewed        |            |
|                                                                                      |                   |      | Notes           |            |
|                                                                                      | Version           | Size | Page 2 Total 5  |            |
|                                                                                      | 1.1               | A4   | Leo Walker 2026 |            |



|           |                   |      |                 |                        |
|-----------|-------------------|------|-----------------|------------------------|
| Page      | CAN_FD            |      | Create at       | 2026-05-30             |
| Schematic | Schematic1_1      |      | Update at       | 2026-06-01             |
| Board     | Module Board 2    |      | Drawn           | Leo Walker             |
|           | can fd usb module |      | Reviewed        |                        |
|           |                   |      | Notes           | Added indication LEDs. |
|           | Version           | Size | Page 3 Total 5  |                        |
|           | 1.1               | A4   | Leo Walker 2026 |                        |



